

CALEB AGUIRRE-LEON

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🐙 <https://github.com/lementknight>

EDUCATION:

Syracuse University

B.S. in Computer Science, Minor Degree in Mathematics

Syracuse, New York

December 2025

PERSONAL SUMMARY:

NYC-based data engineer with production-level experience supporting enterprise clients including Google and IBM, specializing in infrastructure modernization and data pipeline engineering

SKILLS:

Languages: Python, Go, SQL (T-SQL), Java, C, HTML, CSS, JavaScript, Latex, Markdown.

Technologies: AWS (S3, SQS), Snowflake, Apache Airflow, Microsoft SQL Server, Windows Remote Desktop Services, Poetry, Nox, Google Cloud Platform, Google Firebase, GitHub, GitHub Actions, Git, Bazel, Flask, Jinja, MySQL, Nginx, Prometheus, Grafana.

RELEVANT WORK EXPERIENCE:

Data Engineer, Attention Arc (LWDirect), Irvington, New York

March 2026 – Present

- Re-architected the nightly Nielsen ETL pipeline (1TB+/year vendor feed) powering media planning and activation workflows by consolidating three disjoint legacy extract/transform/load scripts (obfuscated JAR, 1,200-line Perl, 6 ad-hoc SQL Server importers) into a single modular Python codebase with schema validation and sharded inserts, cutting batch runtime from 4 hours to under 1 hour.
- Established data observability across the pipeline via structured logging, Slack-based job alerting, and a CI parity-test suite (GitHub Actions, Nox, Poetry, SQLAlchemy) that caught multiple pre-production regressions, enabling team-wide freshness SLA tracking, self-service job status, and detection of silent data-quality issues that previously bypassed validation.
- Implemented end-to-end source-to-target data lineage mapping each raw vendor file to its destination rows, enabling targeted backfills, surfacing gaps in upstream vendor deliveries, and cutting data-issue triage from hours to minutes; the project's repo template (CI/CD, type-checking, parity tests) was adopted as the team-wide standard, and I now lead the same modernization across additional production data pipelines.

Site Reliability Engineering Intern, Google, New York, NY

May 2024 – August 2024

- Improved Monarch monitoring system release velocity by 80% through implementing granular software release freeze controls, reducing deployment downtime and enabling more efficient rollout management for Google's planet-scale infrastructure.
- Designed and implemented binary-to-subsystem relationship modeling system for 30+ production binaries, improving deployment precision by 100% and reducing average blast radius during release freezes by 80%.
- Engineered Go and Bazel-based automated configuration generation tool, reducing manual overhead by 60% and facilitating seamless integration with Google's rollout infrastructure; additionally, created comprehensive documentation to ensure widespread team adoption.

Quantum Open-Source Software Intern, Qubit by Qubit and IBM Quantum, Manhattan, New York

June 2022 – August 2022

- Engineered CLI tool for Qiskit Red Queen Benchmarking Suite, boosting benchmarking coverage by 80% and optimizing quantum circuits with Pylint, Black, and Pytest, benefiting 100+ IBM Qiskit developers
- Executed 10 comprehensive code reviews for IBM's Qiskit Red Queen Benchmarking Suite, elevating code quality by 75% through detailed analysis of algorithmic efficiency, code structure, and adherence to quantum computing best practices.
- Presented quantum computing research findings to 30 program participants and leadership, achieving a perfect 10/10 average feedback score for clarity and content, while enhancing program-wide understanding of advanced quantum concepts.

LEADERSHIP/ACTIVITIES:

Co-Sponsorship Chair, CuseHacks (Innovate Orange), Syracuse, New York

January 2024 – May 2025

- Spearheaded fundraising efforts for CuseHacks' bi-annual hackathon, securing \$7,500 through targeted corporate outreach and leveraging Notion to streamline agreement processes and deadline management.
- Led weekly planning meetings with 20 e-board members, overseeing logistics for bi-annual hackathons and weekly workshops, resulting in 80% increase in event attendance and 75% improvement in participant satisfaction scores.

President, Syracuse University Dean's Advisory Panel, Syracuse, New York

August 2023 – May 2025

- Led 9-member panel, managing 3 committees and executing weekly community outreach events, resulting in 40% increase in student engagement and implementation of 5 new initiatives to enhance engineering student experience.
- Gathered and presented student body concerns to the Dean of Engineering monthly, leading to the implementation of 2 new policies and 5 improvements in student organization resources, benefiting 90% of engineering students.